

Controlling for Additional Variables

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Review of Descriptive Interpretation

- put up summary activity slide from "Descriptive Questions with Linear Regression" lecture
- Write $Y = \beta_0 + \beta_1 x$. Ask for interpretation of β_1 .
- Write $E[Y|x, z] = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 x \cdot z$. Re-write as $E[Y|x, z] = \beta_0 + \beta_2 z + (\beta_1 + \beta_3 z) \cdot x$
Ask class to identify β_1 and β_2 on the plot. As for descriptive interpretation of β_1 .

Mean difference between two observations that have $z = 0$ and a one unit difference in x , assuming the model.

- Ask parameters correspond to the same interpretation at the $z = 1$ level.
- As for descriptive interpretation of β_2 . When might this parameter not be substantively meaningful?
- Draw additive model on board for $Y = \beta_0 + \beta_1 x + \beta_2 z$. Ask class to identify β_1 and β_2 on the plot. Ask for descriptive interpretation of β_1 .

Mean difference between two observations that have the same z value and a one unit difference in x , assuming the model.

- Ask what combination of parameters (for x) could provide the additive interpretation for the interactive model.
- Discuss the relationship between standardization and the additive slope.
- Ask class to identify β_1 and β_2 on the plot. As for descriptive interpretation of β_2 .

Mean difference between two observations that have the same x value and a one unit difference in z , assuming the model.

- Ask what combination of parameters (for z) could provide the additive interpretation for the interactive model.

Hierarchical Principle

- wage $\sim$ age, to show that one needs a non-zero intercept with a linear model
- wage $\sim$ age , CD

Polynomials

- wage age
- special case of interactive model
- interpretation difficult. use marginal.

Controlling

- partial (and extra partial) regression plot (Frisch-Waugh-Lovell Theorem)
- works for any number of variables
- what this implies for x and z uncorrelated.
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$$\hat{B}_x = \frac{R_{xy} - R_{xz}R_{yz}}{1 - R_{xz}^2} \left(\frac{s_y}{s_x} \right)$$

If $R_{xz} = 0$ then

$$\hat{B}_x = \frac{R_{xy} - 0 \cdot R_{yz}}{1 - 0^2} \left(\frac{s_y}{s_x} \right) = R_{xy} \left(\frac{s_y}{s_x} \right)$$

This is the simple linear regression estimator for the slope of a regression of y on x only.

OLS Matrix Representation

- $y = X\beta + \epsilon$, with Epstein Mershon data
- special case of constant
- constant with 2 observations 3/5 and 5/5. (both views)
- LS is projection, projection implies orthogonal to all X
- Derive Matrix LS solution ($X'(y - X\hat{\beta}) = 0$)
- Special case with constant
- example with two observations
- example with x centered (example with x and z centered for problem set)
- orthogonal x and z centered.